I/O Systems

15-213: Introduction to Computer Systems
16th Lecture, October 19th, 2017

Instructor:

Randy Bryant

Today

- **I/O Systems**
- Unix I/O
- Metadata, sharing, and redirection
- Standard I/O
- Closing remarks

Overview

- **A computer's job is to process data**
  - Computer (CPU, cache, and memory)
  - Move data into and out of a system (between I/O devices and memory)
- **Challenges with I/O devices**
  - Different categories: storage, networking, displays, etc.
  - Large number of device drivers to support
  - Device driver run in kernel mode and can crash systems

Overview (Cont.)

- **I/O management is a major component of operating system**
  - Important aspect of computer operation
  - I/O devices vary greatly
  - Various methods to control them
  - Performance management
  - New types of devices frequent
- **Ports, busses, device controllers connect to various devices**
- **Device drivers encapsulate device details**
  - Present uniform device-access interface to I/O subsystem

操作系统在程序执行过程中的作用

- Shell进程生成子进程，子进程调用execve系统调用启动加载器，以装入Hello程序，最后跳转到第一条指令执行
- 在Hello程序执行过程中，Hello本身不会直接访问键盘、显示器、磁盘和主存储器等硬件资源，而是依靠OS提供的服务来间接访问。

  例如，利用printf()函数最终调出内核服务程序访问硬件。
- 操作系统是在应用程序和硬件之间的中间软件层。
- 操作系统的两个主要的作用：
  - 硬件资源管理，以达到以下两个目的：
    - 统筹安排和调度硬件资源，以防止硬件资源被用户程序滥用
    - 对于广泛使用的复杂低级设备，为用户程序提供一个简单一致的使用接口
  - 为用户（最终用户、用户程序）使用系统提供一个操作接口

I/O子系统

各类用户的I/O请求需要通过某种方式传给OS：

- **最终用户：键盘、鼠标通过操作界面传递给OS**
- **用户程序：通过函数（高级语言）转换为系统调用传递给OS**

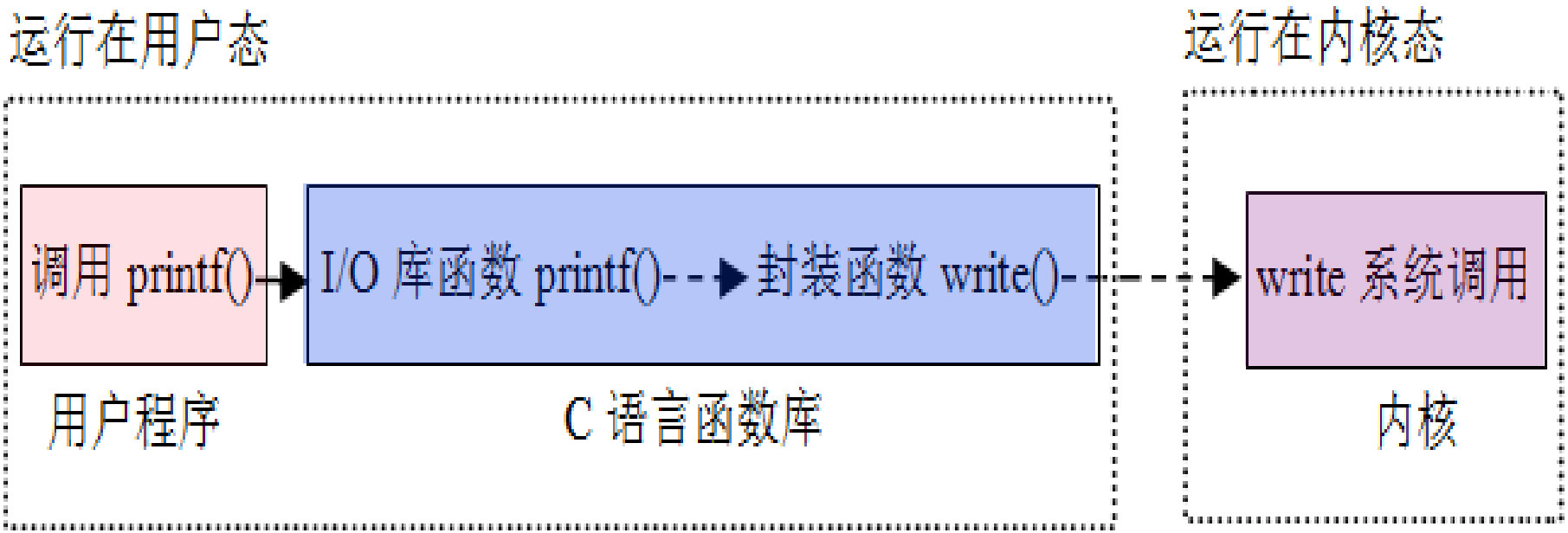
大部分I/O软件都属于操作系统内核态程序，最初的I/O请求在用户程序中提出。

用户程序、C函数和内核

- 用户程序总是通过某种I/O函数或I/O操作符请求I/O操作。

  例如，读一个磁盘文件记录时，可调用C标准I/O库函数fread()，也可直接调用系统调用封装函数read()来提出I/O请求。不管是C库函数、API函数还是系统调用封装函数，最终都通过操作系统内核提供的系统调用来实现I/O。

  printf()函数的调用过程如下：



Today

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- **Unix I/O**
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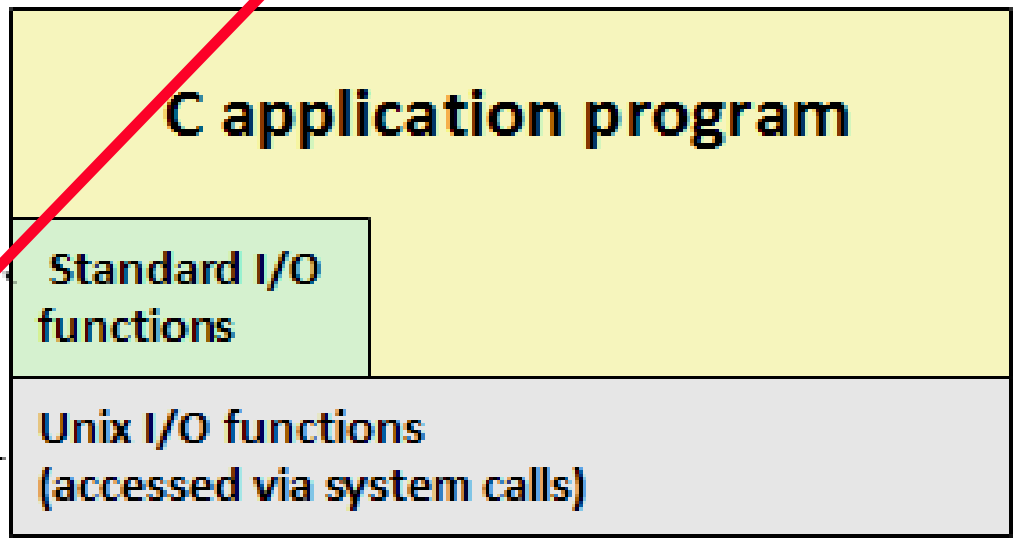
UNIX/Linux的I/O

- 用户程序可通过调用特定的I/O函数的方式提出I/O请求。
- 在UNIX/Linux系统中，可以是C标准I/O库函数或系统调用的封装函数，前者如文件I/O函数fopen()、fread()、fwrite()和fclose()或控制台I/O函数printf()、putc()、scanf()和getc()等；后者如open()、read()、write()和close()等。
- 标准I/O库函数比系统调用封装函数抽象层次高，后者属于系统级I/O函数。与系统提供的API函数一样，前者是基于后者实现的。

```
fopen   fdopen
fread   fwrite
fscanf  fprintf
sscanf  sprintf
fgets   fputs
fflush  fseek
fclose
```

```
open    read
write   lseek
stat    close
```



Unix I/O Overview

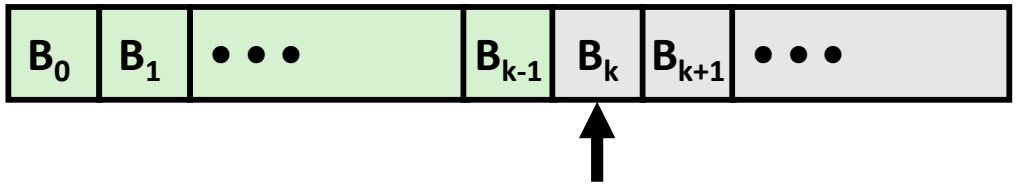
- **A Linux *file* is a sequence of *m* bytes:**
  - $B_0, B_1, \ldots, B_k, \ldots, B_{m-1}$

- **Cool fact: All I/O devices are represented as files:**
  - `/dev/sda2` (`/usr` disk partition)
  - `/dev/tty2` (terminal)

- **Even the kernel is represented as a file:**
  - `/boot/vmlinuz-3.13.0-55-generic` (kernel image)
  - `/proc` (kernel data structures)

Unix I/O Overview

- **Elegant mapping of files to devices allows kernel to export simple interface called *Unix I/O:***
  - Opening and closing files
    - `open()` and `close()`
  - Reading and writing a file
    - `read()` and `write()`
  - Changing the ***current file position*** (seek)
    - indicates next offset into file to read or write
    - `lseek()`



Current file position = k

File Types

- **Each file has a *type* indicating its role in the system**
  - *Regular file:* Contains arbitrary data
  - *Directory:* Index for a related group of files
  - *Socket:* For communicating with a process on another machine

- **Other file types beyond our scope**
  - *Named pipes (FIFOs)*
  - *Symbolic links*
  - ***Character*** *and block devices*

Regular Files

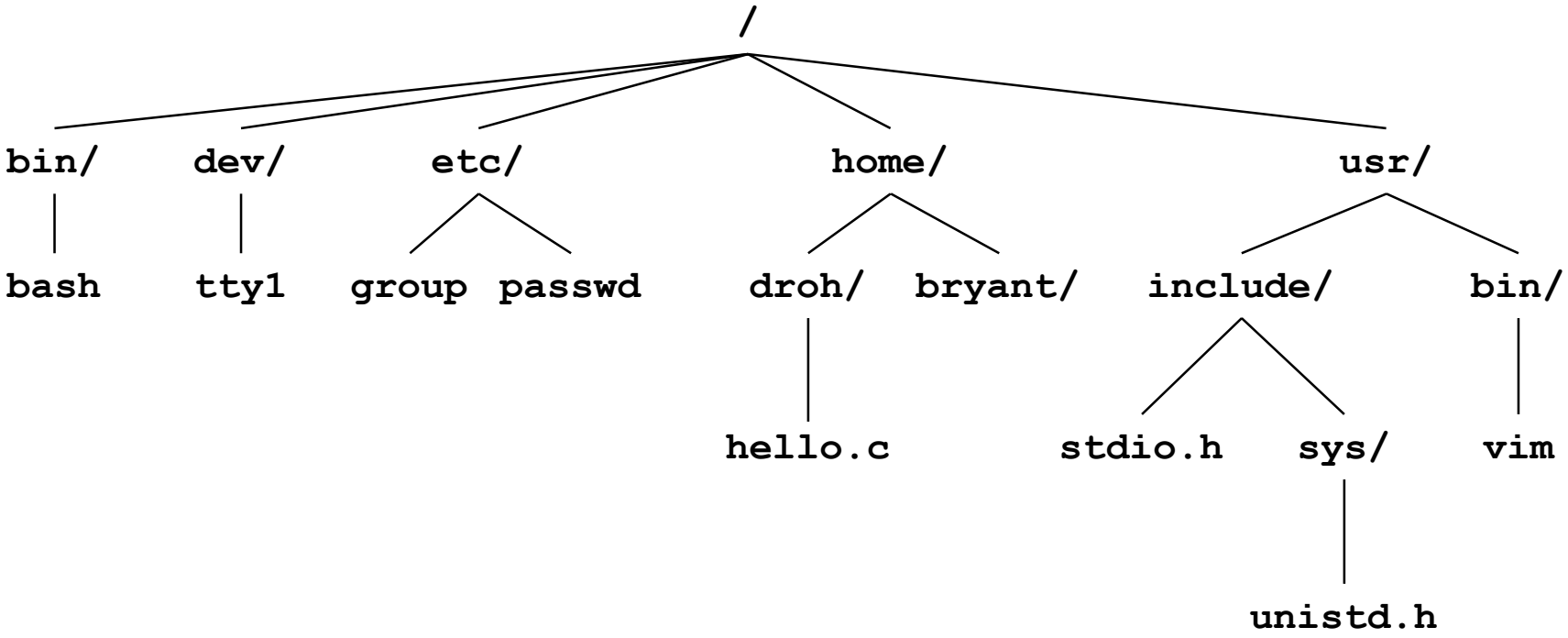
- **A regular file contains arbitrary data**
- **Applications often distinguish between *text files* and *binary files***
  - Text files are regular files with only ASCII or Unicode characters
  - Binary files are everything else
    - e.g., object files, JPEG images
  - **Kernel doesn't know the difference!**
- **Text file is sequence of *text lines***
  - Text line is sequence of chars terminated by *newline char* (**'`\n`'**)
    - Newline is **`0xa`**, same as ASCII line feed character (LF)
- **End of line (EOL) indicators in other systems**
  - Linux and Mac OS: **'`\n`'** (**`0xa`**)
    - line feed (LF)
  - Windows and Internet protocols: **'`\r\n`'** (**`0xd 0xa`**)
    - Carriage return (CR) followed by line feed (LF)

Directories

- **Directory consists of an array of *links***
  - Each link maps a *filename* to a file
- **Each directory contains at least two entries**
  - `.` (dot) is a link to itself
  - `..` (dot dot) is a link to *the parent directory* in the *directory hierarchy* (next slide)
- **Commands for manipulating directories**
  - **`mkdir`**: create empty directory
  - **`ls`**: view directory contents
  - **`rmdir`**: delete empty directory

Directory Hierarchy

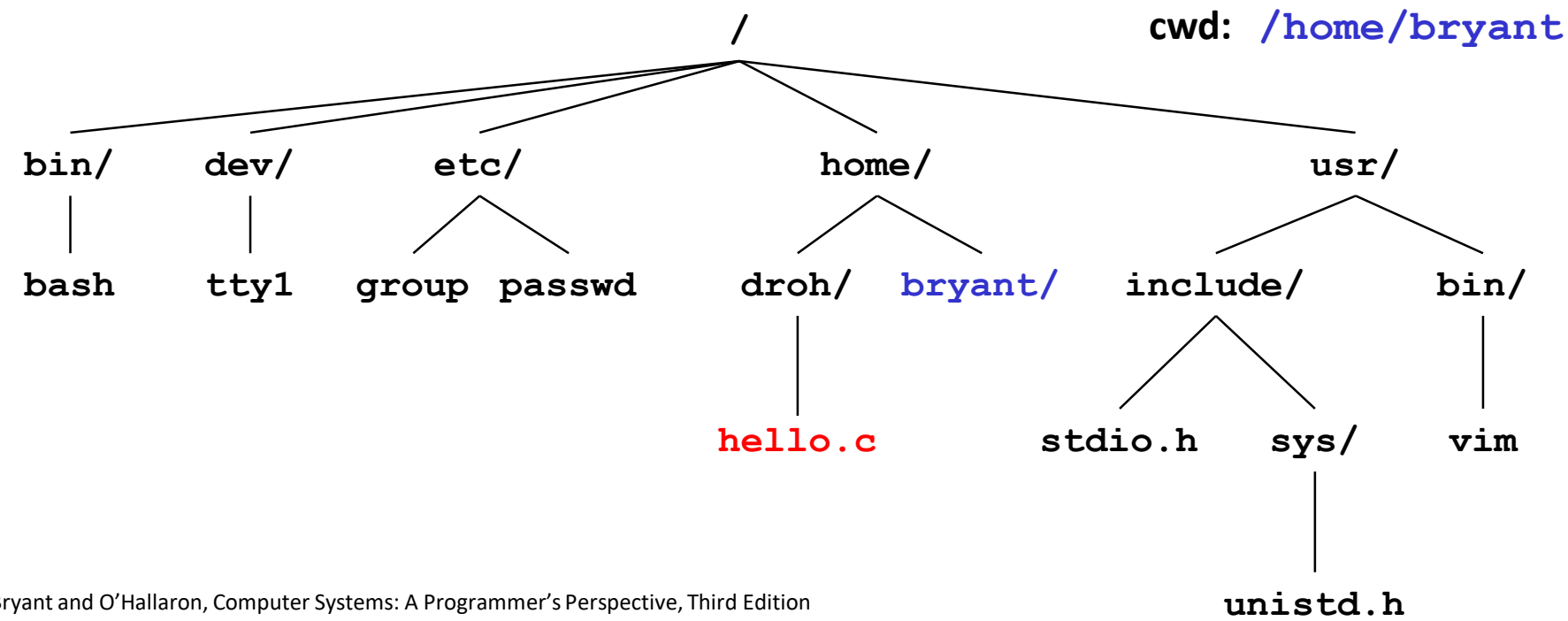
- **All files are organized as a hierarchy anchored by root directory named / (slash)**



- **Kernel maintains *current working directory (cwd)* for each process**
  - Modified using the `cd` command

Pathnames

- **Locations of files in the hierarchy denoted by *pathnames***
  - *Absolute pathname* starts with '/' and denotes path from root
    - **`/home/droh/hello.c`**
  - *Relative pathname* denotes path from current working directory
    - **`../droh/hello.c`**



Opening Files

- **Opening a file informs the kernel that you are getting ready to access that file**

```
int fd;   /* file descriptor */

if ((fd = open("/etc/hosts", O_RDONLY)) < 0) {
   perror("open");
   exit(1);
}
```

- **Returns a small identifying integer *file descriptor***
  - `fd == -1` indicates that an error occurred
- **Each process created by a Linux shell begins life with three open files associated with a terminal:**
  - 0: standard input (stdin)
  - 1: standard output (stdout)
  - 2: standard error (stderr)

Closing Files

- Closing a file informs the kernel that you are finished accessing that file

```
int fd;     /* file descriptor */
int retval; /* return value */

if ((retval = close(fd)) < 0) {
   perror("close");
   exit(1);
}
```

- Closing an already closed file is a recipe for disaster in threaded programs
- Moral: Always check return codes, even for seemingly benign functions such as `close()`

Reading Files

- **Reading a file copies bytes from the current file position to memory, and then updates file position**

```
char buf[512];
int fd;        /* file descriptor */
int nbytes;    /* number of bytes read */

/* Open file fd ...  */
/* Then read up to 512 bytes from file fd */
if ((nbytes = read(fd, buf, sizeof(buf))) < 0) {
   perror("read");
   exit(1);
}
```

- **Returns number of bytes read from file `fd` into `buf`**
  - Return type `ssize_t` is signed integer
  - `nbytes < 0` indicates that an error occurred
  - ***Short counts*** (`nbytes < sizeof(buf)`) are possible and are not errors!

Writing Files

- **Writing a file copies bytes from memory to the current file position, and then updates current file position**

```
char buf[512];
int fd;        /* file descriptor */
int nbytes;    /* number of bytes read */

/* Open the file fd ... */
/* Then write up to 512 bytes from buf to file fd */
if ((nbytes = write(fd, buf, sizeof(buf)) < 0) {
   perror("write");
   exit(1);
}
```

- **Returns number of bytes written from `buf` to file `fd`**
  - `nbytes < 0` indicates that an error occurred
  - As with reads, short counts are possible and are not errors!

Simple Unix I/O example

- **Copying stdin to stdout, one byte at a time**

```
#include "csapp.h"

int main(void)
{
    char c;

    while(Read(STDIN_FILENO, &c, 1) != 0)
        Write(STDOUT_FILENO, &c, 1);
    exit(0);
}
```

On Short Counts

- **Short counts can occur in these situations:**
  - Encountering (end-of-file) EOF on reads
  - Reading text lines from a terminal
  - Reading and writing network sockets
- **Short counts never occur in these situations:**
  - Reading from disk files (except for EOF)
  - Writing to disk files
- **Best practice is to always allow for short counts.**

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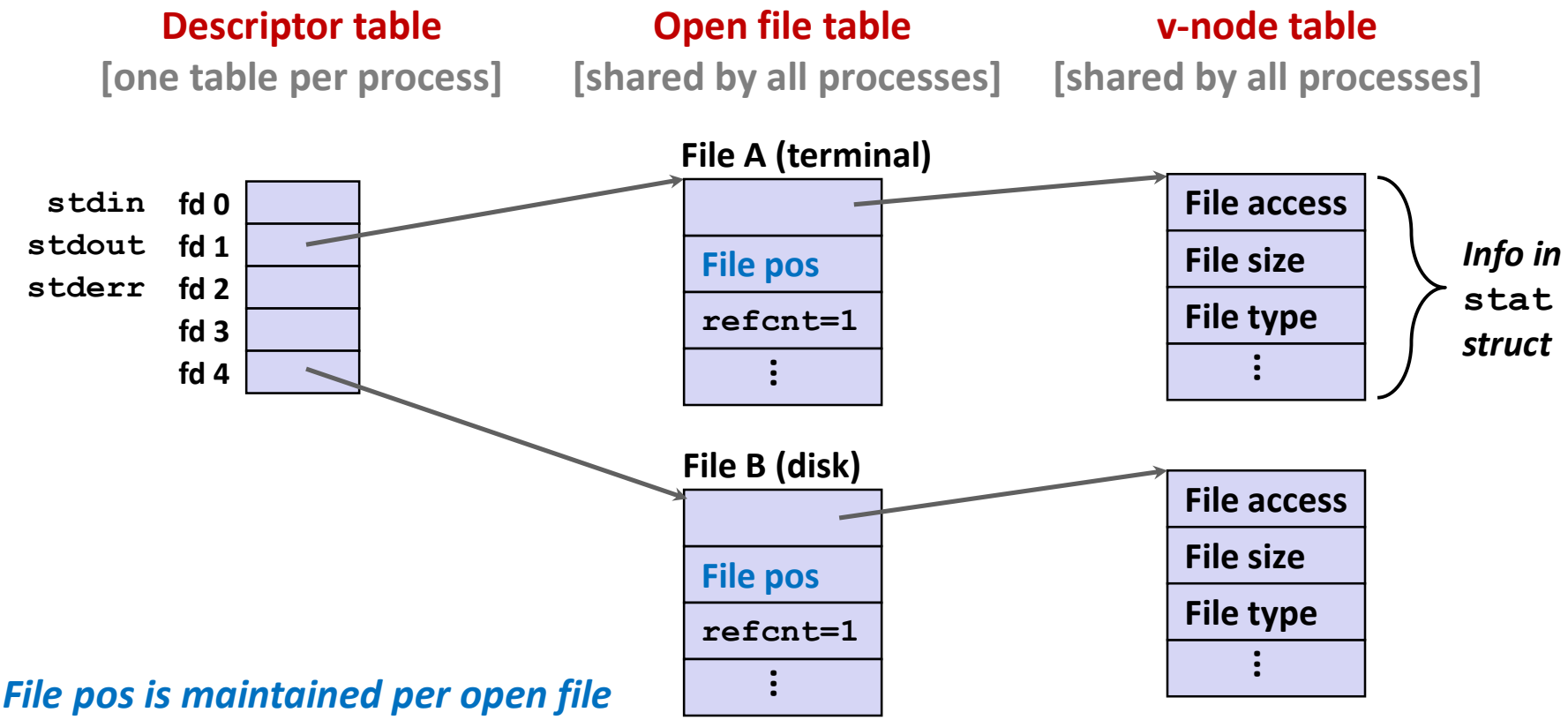
File Metadata

- ***Metadata*** **is data about data, in this case file data**
- **Per-file metadata maintained by kernel**
  - accessed by users with the **`stat`** and **`fstat`** functions

```
/* Metadata returned by the stat and fstat functions */
struct stat {
    dev_t         st_dev;      /* Device */
    ino_t         st_ino;      /* inode */
    mode_t        st_mode;     /* Protection and file type */
    nlink_t       st_nlink;    /* Number of hard links */
    uid_t         st_uid;      /* User ID of owner */
    gid_t         st_gid;      /* Group ID of owner */
    dev_t         st_rdev;     /* Device type (if inode device) */
    off_t         st_size;     /* Total size, in bytes */
    unsigned long st_blksize;  /* Blocksize for filesystem I/O */
    unsigned long st_blocks;   /* Number of blocks allocated */
    time_t        st_atime;    /* Time of last access */
    time_t        st_mtime;    /* Time of last modification */
    time_t        st_ctime;    /* Time of last change */
};
```

How the Unix Kernel Represents Open Files

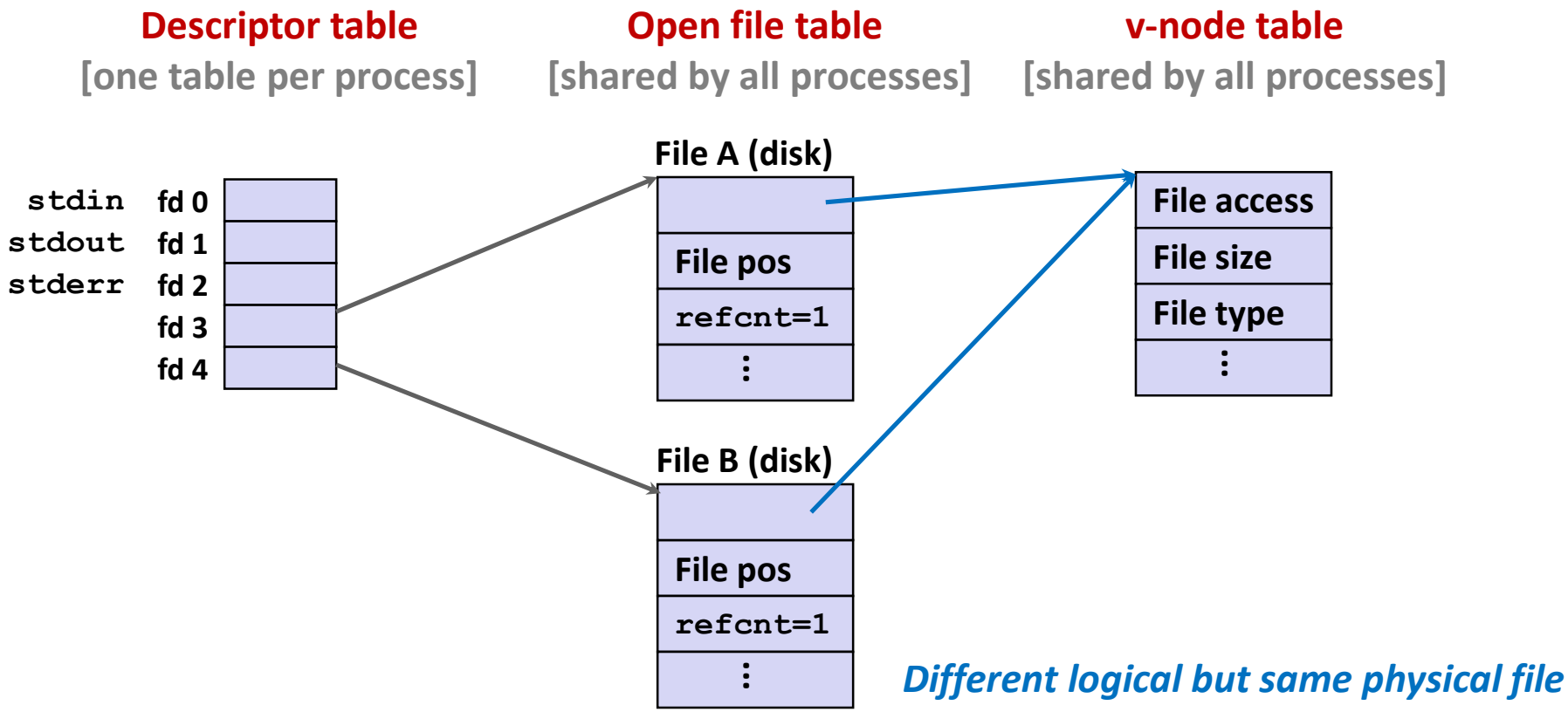
- Two descriptors referencing two distinct open files. Descriptor 1 (stdout) points to terminal, and descriptor 4 points to open disk file



File pos is maintained per open file

File Sharing

- **Two distinct descriptors sharing the same disk file through two distinct open file table entries**
  - E.g., Calling **`open`** twice with the same **`filename`** argument



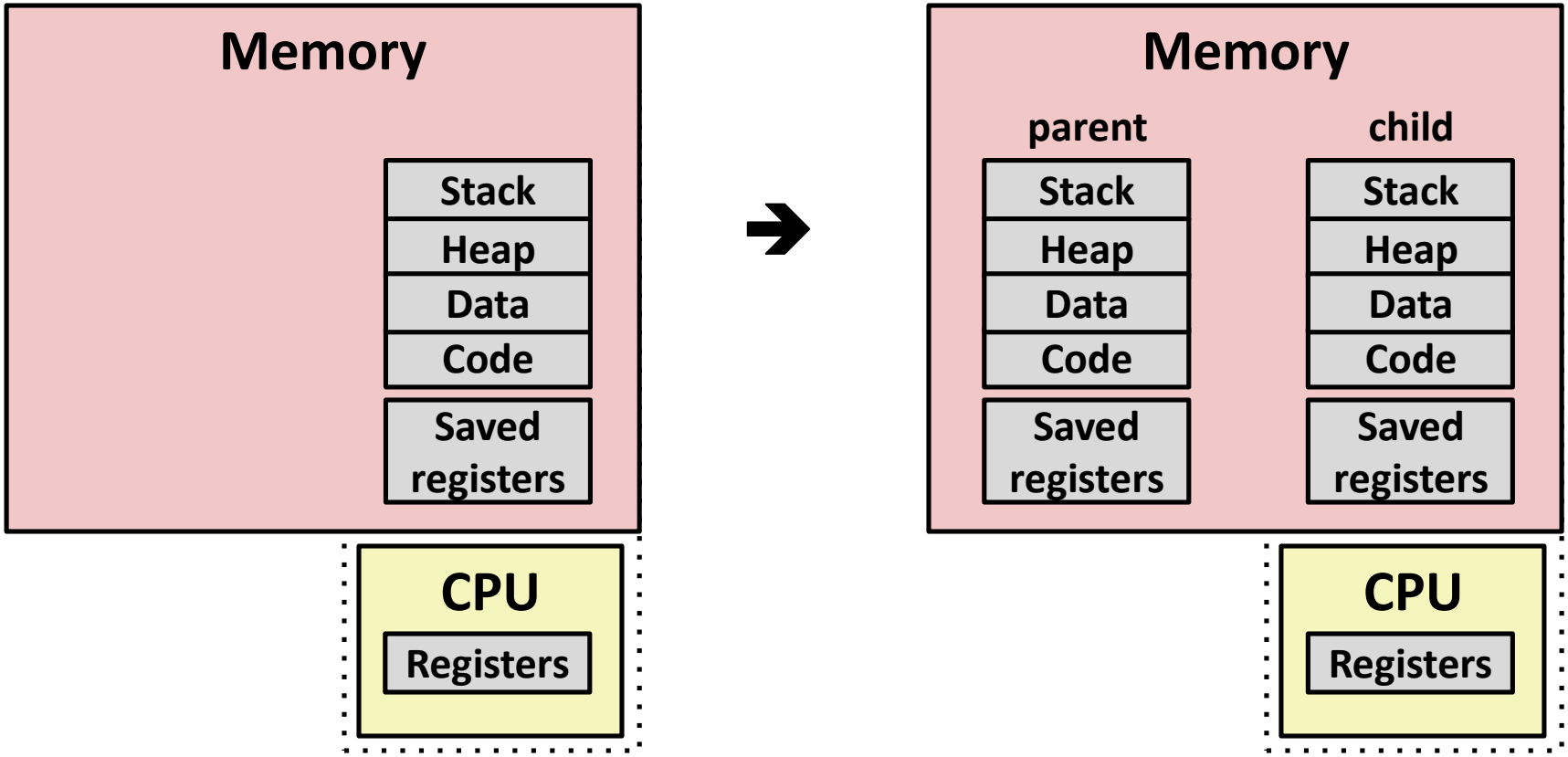
Creating Processes

- ***Parent process*** **creates a new running** ***child process*** **by calling `fork`**

- **`int fork(void)`**
  - Returns 0 to the child process, child's PID to parent process
  - Child is *almost* identical to parent:
    - Child get an identical (but separate) copy of the parent's virtual address space.
    - Child gets identical copies of the parent's open file descriptors
    - Child has a different PID than the parent

- **`fork` is interesting (and often confusing) because it is called *once* but returns *twice***

Conceptual View of `fork`



- **Make complete copy of execution state**
  - Designate one as parent and one as child
  - Resume execution of parent or child

fork Example

```
int main(int argc, char** argv)
{
    pid_t pid;
    int x = 1;

    pid = Fork();
    if (pid == 0) {  /* Child */
        printf("child : x=%d\n", ++x);
        return 0;
    }

    /* Parent */
    printf("parent: x=%d\n", --x);
    return 0;
}
```

fork.c

- Call once, return twice
- Concurrent execution
  - Can't predict execution order of parent and child

```
linux> ./fork
parent: x=0
child : x=2
```

```
linux> ./fork
child : x=2
parent: x=0
```

```
linux> ./fork
parent: x=0
child : x=2
```

```
linux> ./fork
parent: x=0
child : x=2
```

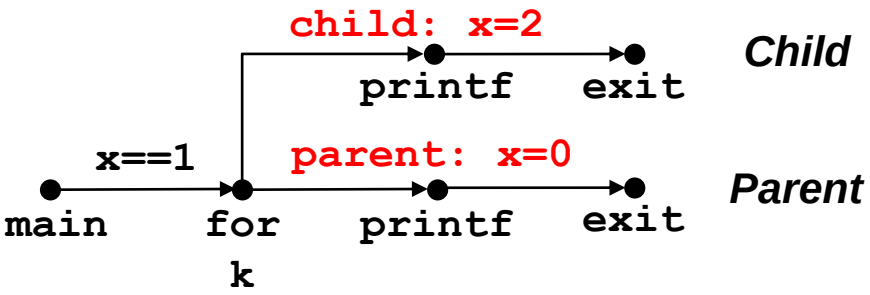
Modeling `fork` with Process Graphs

- **A *process graph* is a useful tool for capturing the partial ordering of statements in a concurrent program:**
  - Each vertex is the execution of a statement
  - a -> b means `a` happens before b
  - Edges can be labeled with current value of variables
  - `printf` vertices can be labeled with output
  - Each graph begins with a vertex with no inedges
- **Any *topological sort* of the graph corresponds to a feasible total ordering.**
  - Total ordering of vertices where all edges point from left to right

Process Graph Example

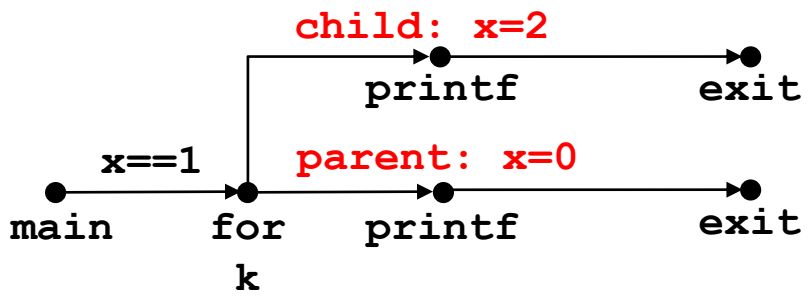
```
int main(int argc, char** argv)
{
    pid_t pid;
    int x = 1;

    pid = Fork();
    if (pid == 0) {  /* Child */
        printf("child : x=%d\n", ++x);
        return 0;
    }

    /* Parent */
    printf("parent: x=%d\n", --x);
    return 0;
}
```

fork.c

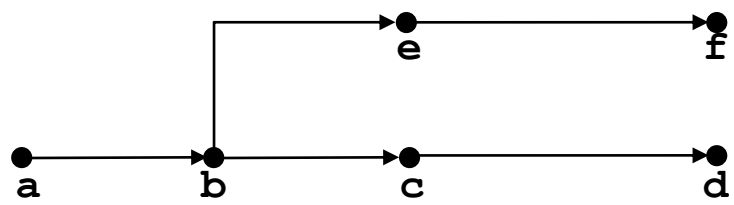


Interpreting Process Graphs

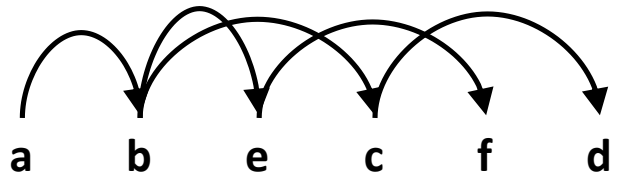
- **Original graph:**



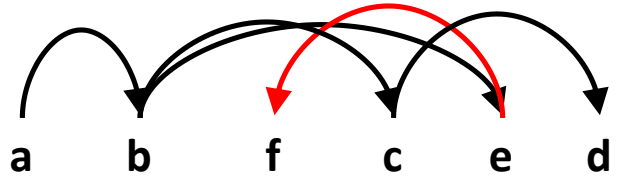
- **Relabled graph:**



Feasible total ordering:

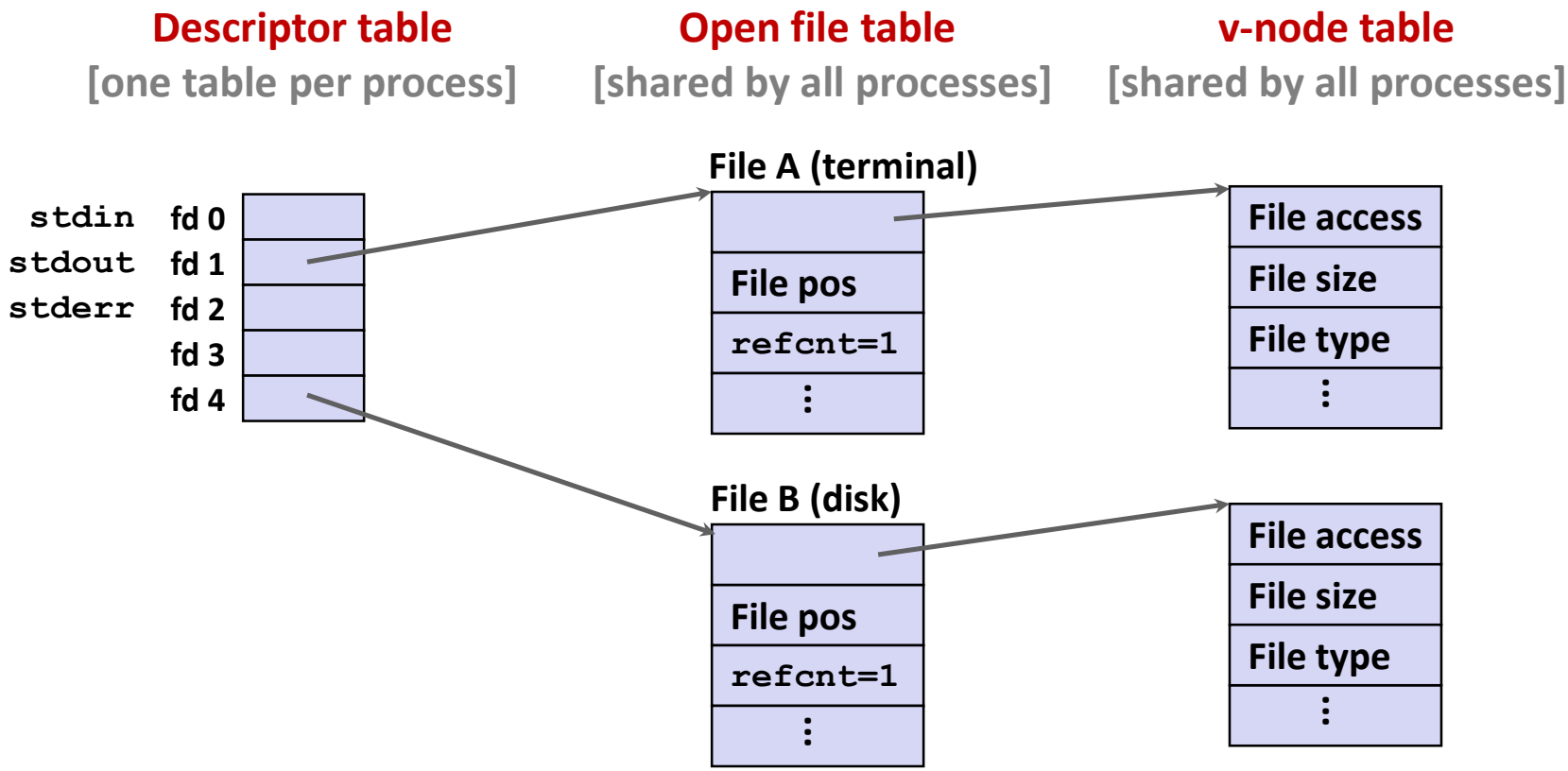


Infeasible total ordering:



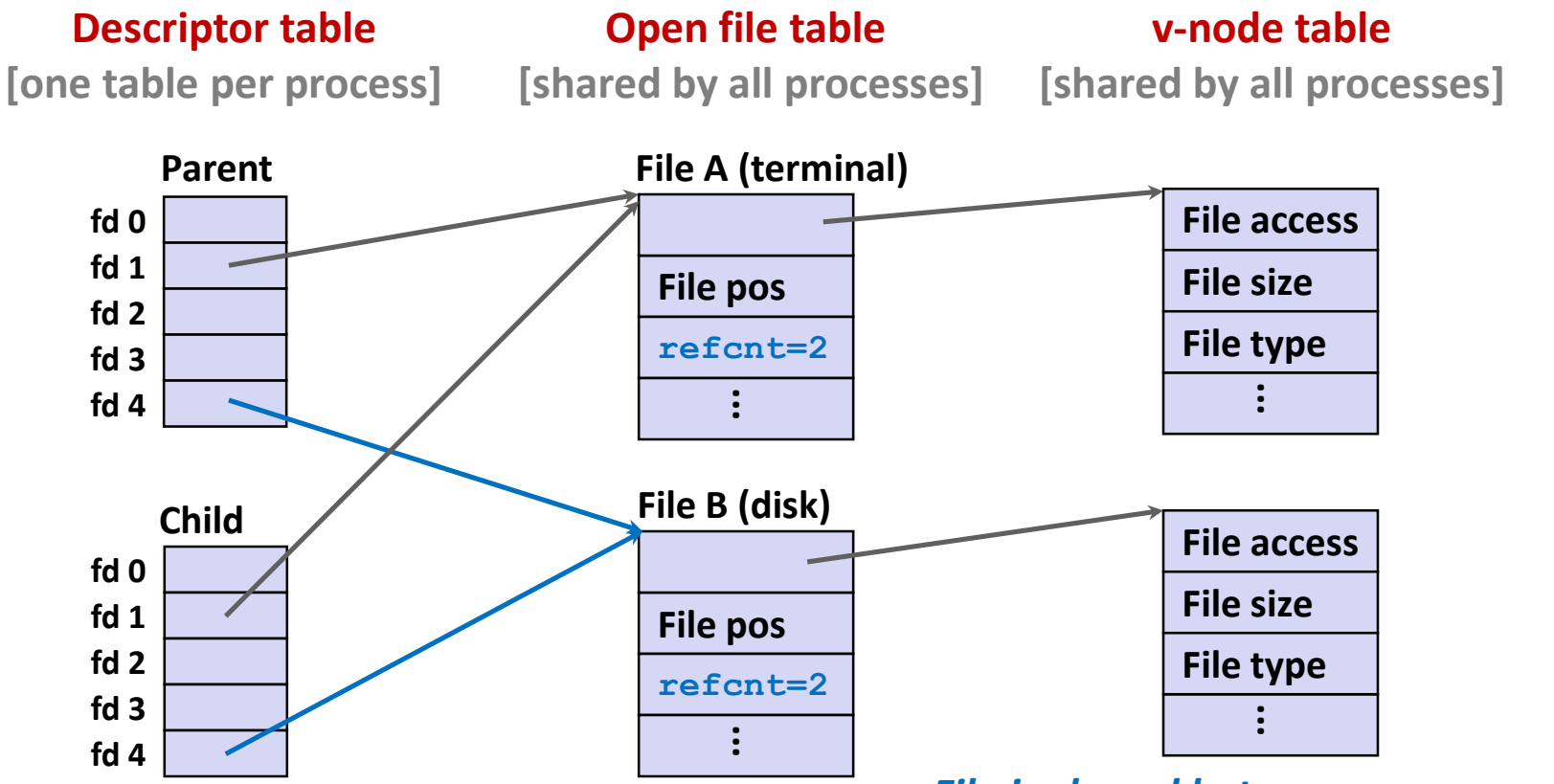
How Processes Share Files: `fork`

- **A child process inherits its parent's open files**
  - Note: situation unchanged by **`exec`** functions (use **`fcntl`** to change)
- ***Before*** **`fork`** **call:**



How Processes Share Files: `fork`

- **A child process inherits its parent's open files**
- ***After* `fork`:**
  - Child's table same as parent's, and +1 to each refcnt



File is shared between processes

I/O Redirection

- **Question: How does a shell implement I/O redirection?**

```
linux> ls > foo.txt
```

- **Answer: By calling the `dup2(oldfd, newfd)` function**
  - Copies (per-process) descriptor table entry `oldfd` to entry `newfd`

Descriptor table *before* `dup2(4,1)`

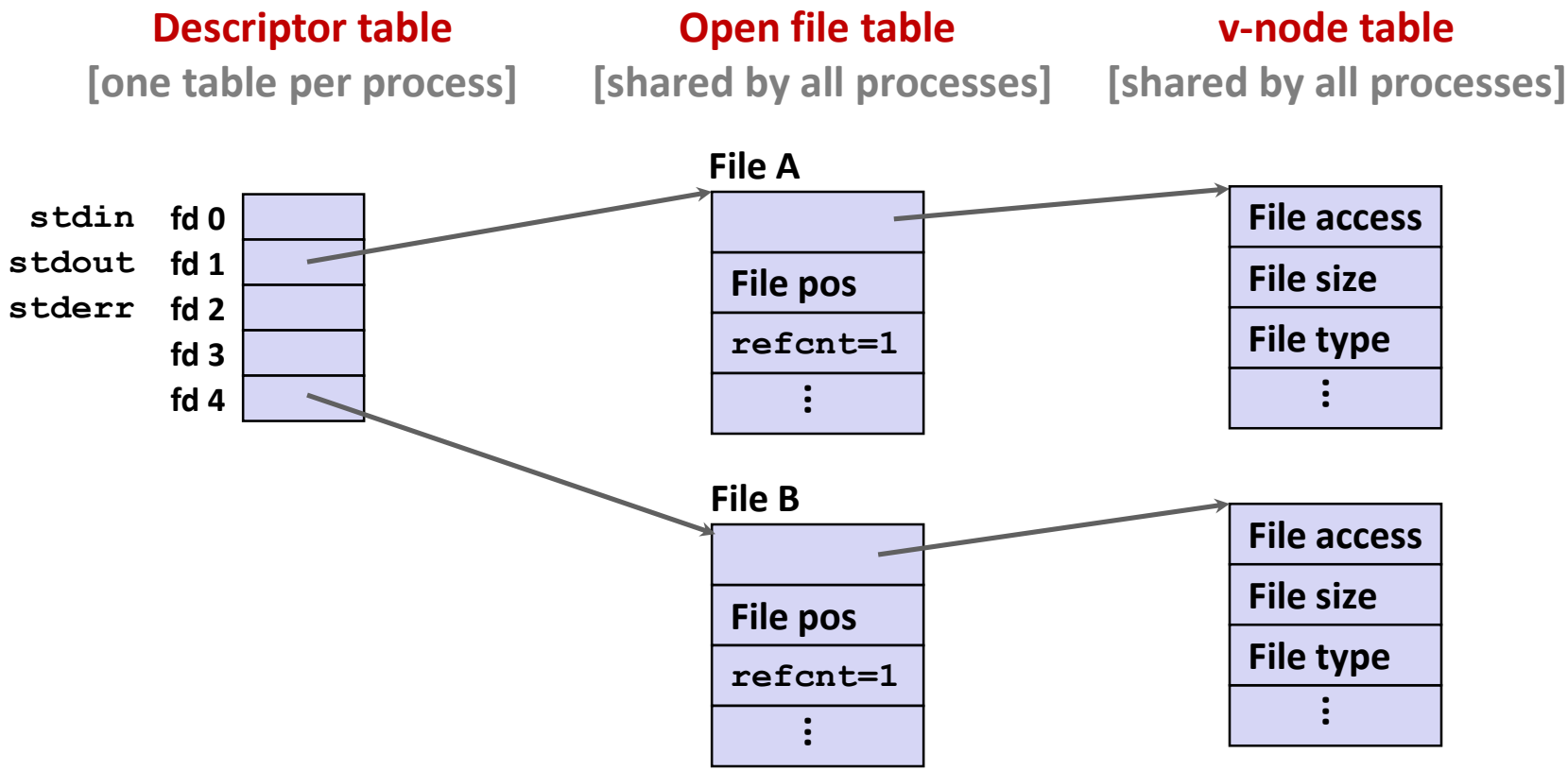
fd 0	
fd 1	a
fd 2	
fd 3	
fd 4	b

Descriptor table *after* `dup2(4,1)`

fd 0	
fd 1	b
fd 2	
fd 3	
fd 4	b

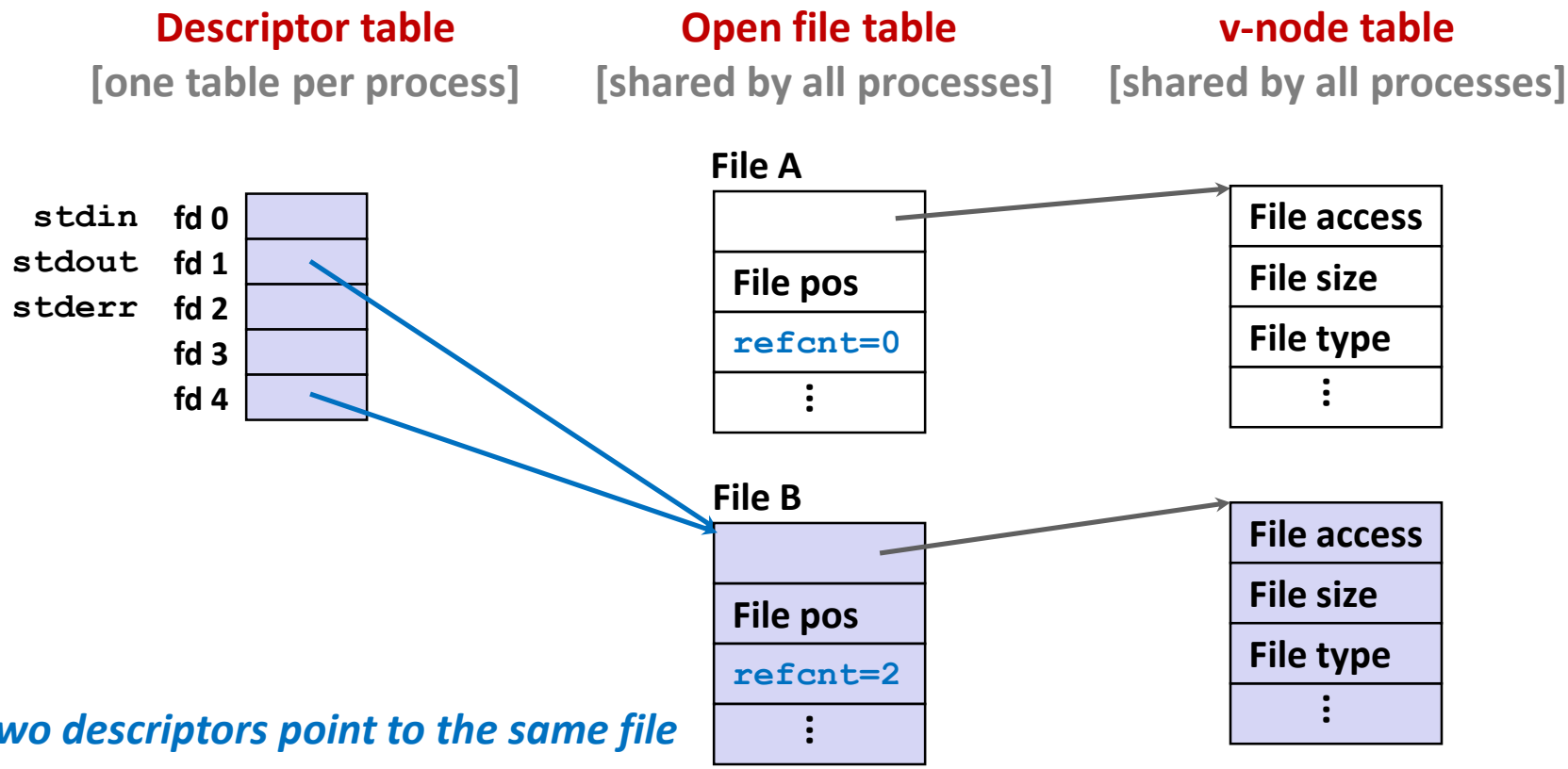
I/O Redirection Example

- **Step #1: open file to which stdout should be redirected**
  - Happens in child executing shell code, before **`exec`**



I/O Redirection Example (cont.)

- **Step #2: call `dup2(4,1)`**
  - cause fd=1 (stdout) to refer to disk file pointed at by fd=4



Two descriptors point to the same file

Warm-Up: I/O and Redirection Example

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1, fd2, fd3;
    char c1, c2, c3;
    char *fname = argv[1];
    fd1 = Open(fname, O_RDONLY, 0);
    fd2 = Open(fname, O_RDONLY, 0);
    fd3 = Open(fname, O_RDONLY, 0);
    Dup2(fd2, fd3);
    Read(fd1, &c1, 1);
    Read(fd2, &c2, 1);
    Read(fd3, &c3, 1);
    printf("c1 = %c, c2 = %c, c3 = %c\n", c1, c2, c3);
    return 0;
}
```

ffiles1.c

- **What would this program print for file containing "`abcde`"?**

Warm-Up: I/O and Redirection Example

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1, fd2, fd3;
    char c1, c2, c3;
    char *fname = argv[1];
    fd1 = Open(fname, O_RDONLY, 0);
    fd2 = Open(fname, O_RDONLY, 0);
    fd3 = Open(fname, O_RDONLY, 0);
    Dup2(fd2, fd3);
    Read(fd1, &c1, 1);
    Read(fd2, &c2, 1);
    Read(fd3, &c3, 1);
    printf("c1 = %c, c2 = %c, c3 = %c\n", c1, c2, c3);
    return 0;
}
```

ffiles1.c

```
c1 = a, c2 = a, c3 = b
```

```
dup2(oldfd, newfd)
```

- What would this program print for file containing "abcde"?

Master Class: Process Control and I/O

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1;
    int s = getpid() & 0x1;
    char c1, c2;
    char *fname = argv[1];
    fd1 = Open(fname, O_RDONLY, 0);
    Read(fd1, &c1, 1);
    if (fork()) { /* Parent */
        sleep(s);
        Read(fd1, &c2, 1);
        printf("Parent: c1 = %c, c2 = %c\n", c1, c2);
    } else { /* Child */
        sleep(1-s);
        Read(fd1, &c2, 1);
        printf("Child: c1 = %c, c2 = %c\n", c1, c2);
    }
    return 0;
}
```

ffiles2.c

- **What would this program print for file containing "abcde"?**

Master Class: Process Control and I/O

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1;
    int s = getpid() & 0x1;
    char c1, c2;
    char *fname = argv[1];
    fd1 = Open(fname, O_RDONLY, 0);
    Read(fd1, &c1, 1);
    if (fork()) { /* Parent */
        sleep(s);
        Read(fd1, &c2, 1);
        printf("Parent: c1 = %c, c2 = %c\n", c1, c2);
    } else { /* Child */
        sleep(1-s);
        Read(fd1, &c2, 1);
        printf("Child: c1 = %c, c2 = %c\n", c1, c2);
    }
    return 0;
}
```

ffiles2.c

```
Child: c1 = a, c2 = b
Parent: c1 = a, c2 = c
```

```
Parent: c1 = a, c2 = b
Child: c1 = a, c2 = c
```

Bonus: Which way does it go?

- **What would this program print for file containing "abcde"?**

Today

- I/O Systems
- Unix I/O
- Metadata, sharing, and redirection
- **Standard I/O**
- Closing remarks

Standard I/O Functions

- **The C standard library (`libc.so`) contains a collection of higher-level *standard I/O* functions**
  - Documented in Appendix B of K&R

- **Examples of standard I/O functions:**
  - Opening and closing files (**`fopen`** and **`fclose`**)
  - Reading and writing bytes (**`fread`** and **`fwrite`**)
  - Reading and writing text lines (**`fgets`** and **`fputs`**)
  - Formatted reading and writing (**`fscanf`** and **`fprintf`**)

用户I/O软件

- 用户进程请求读磁盘文件操作
  - 用户进程使用标准C库函数fread，或Windows API函数ReadFile，或Unix/Linux的系统调用函数read等要求读一个磁盘文件块。
  - 用户程序中涉及I/O操作的函数最终会被转换为一组与具体机器架构相关的指令序列，这里我们将其称为I/O请求指令序列。
  - 每个指令系统中一定有一类陷阱指令（有些机器也称为软中断指令或系统调用指令），主要功能是为操作系统提供灵活的系统调用机制。
  - 在I/O请求指令序列中，具体I/O请求被转换为一条陷阱指令，在陷阱指令前面则是相应的系统调用参数的设置指令。

以hello程序为例说明

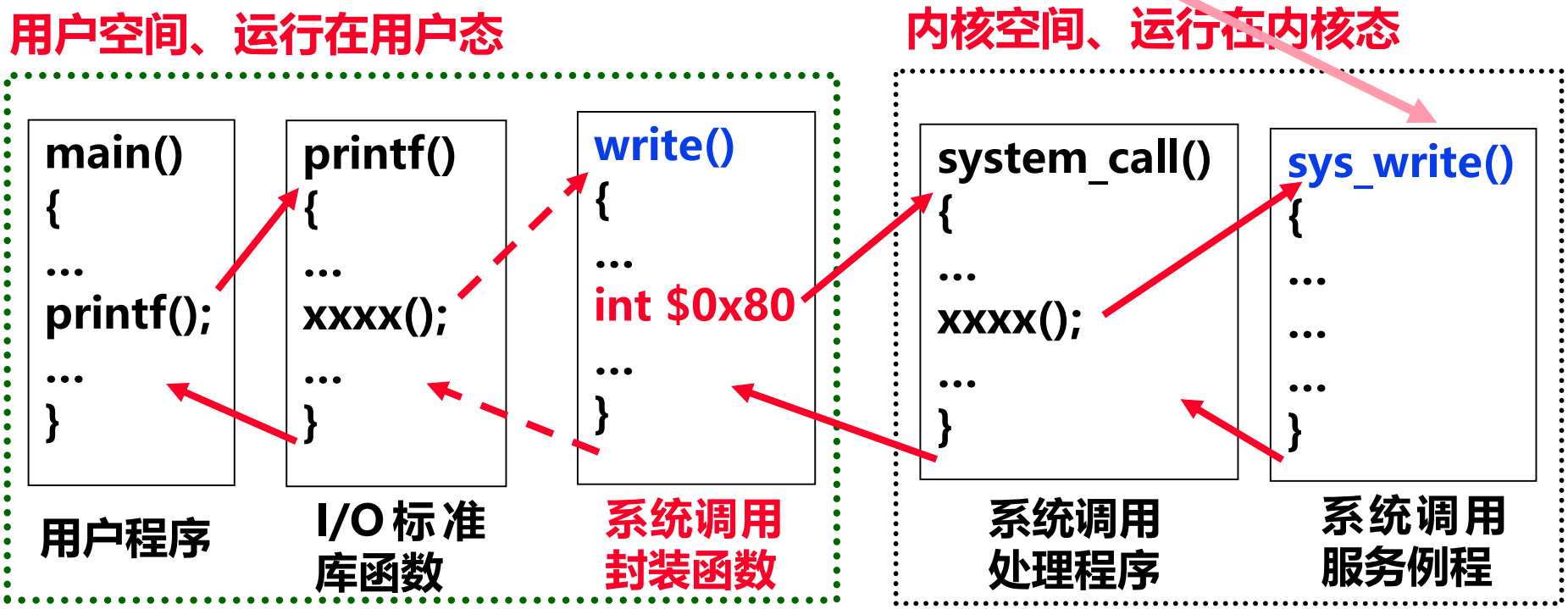
假定以下用户程序对应的进程为p

```
#include <stdio.h>
int main()
{
    printf("hello, world\n");
}
```

sys_write可用三种I/O方式实现：

程序查询、中断 和 DMA

可见：字符串输出最终是由内核中的sys_write系统调用服务例程实现



Standard I/O Streams

- **Standard I/O models open files as *streams***
  - Abstraction for a file descriptor and a buffer in memory

- **C programs begin life with three open streams (defined in `stdio.h`)**
  - **`stdin`** (standard input)
  - **`stdout`** (standard output)
  - **`stderr`** (standard error)

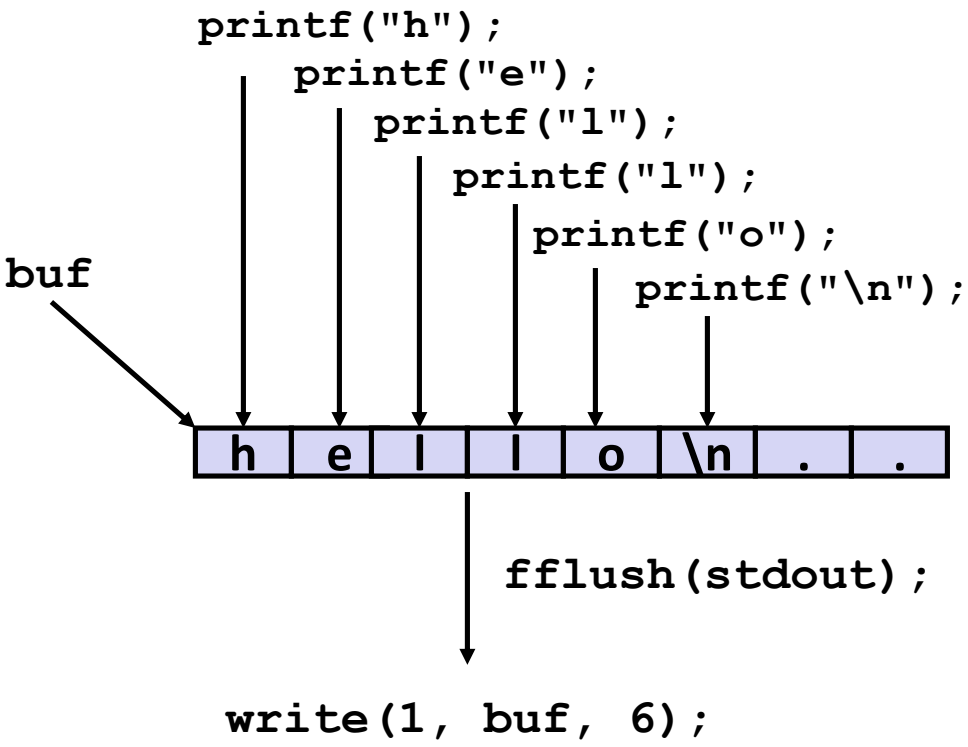
```
#include <stdio.h>
extern FILE *stdin;  /* standard input  (descriptor 0) */
extern FILE *stdout; /* standard output (descriptor 1) */
extern FILE *stderr; /* standard error  (descriptor 2) */

int main() {
    fprintf(stdout, "Hello, world\n");
}
```

Buffered I/O: Motivation

- **Applications often read/write one character at a time**
  - `getc, putc, ungetc`
  - `gets, fgets`
    - Read line of text one character at a time, stopping at newline
- **Implementing as Unix I/O calls expensive**
  - `read` and `write` require Unix kernel calls
    - > 10,000 clock cycles
- **Solution: Buffered read**
  - Use Unix `read` to grab block of bytes
  - User input functions take one byte at a time from buffer
    - Refill buffer when empty

Buffering in Standard I/O

- Standard I/O functions use buffered I/O



- Buffer flushed to output fd on "\n", call to `fflush` or `exit`, or return from `main`.

Standard I/O Buffering in Action

- You can see this buffering in action for yourself, using the always fascinating Linux `strace` program:

```
#include <stdio.h>

int main()
{
    printf("h");
    printf("e");
    printf("l");
    printf("l");
    printf("o");
    printf("\n");
    fflush(stdout);
    exit(0);
}
```

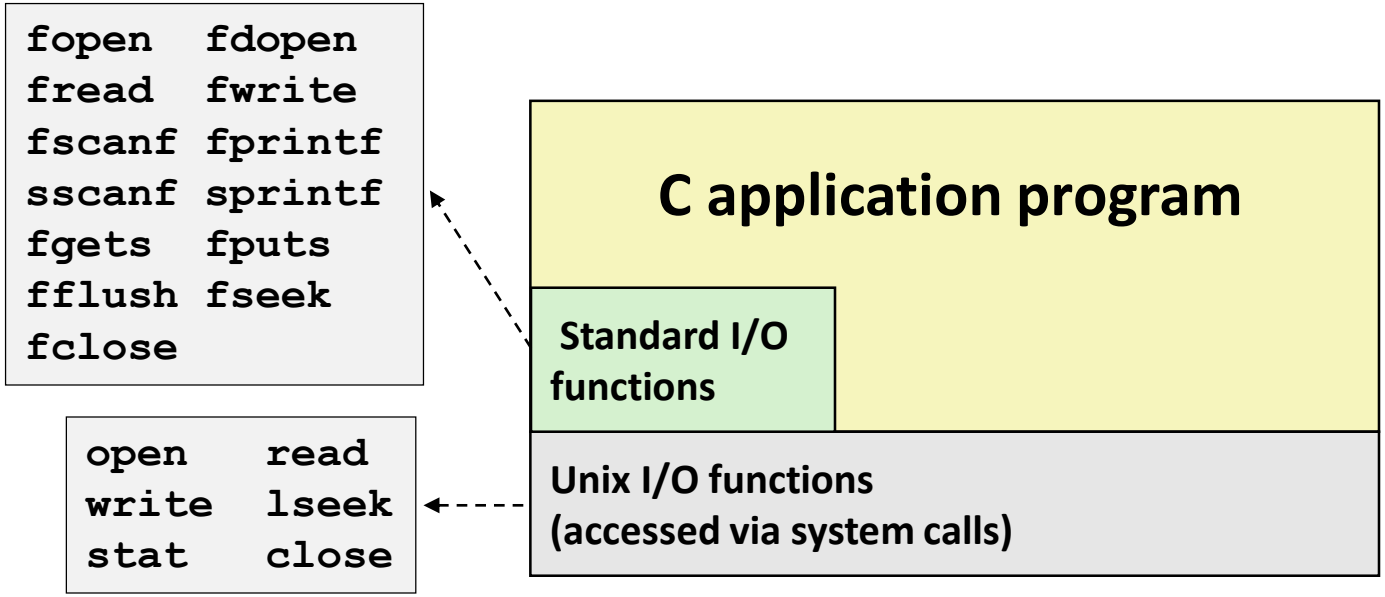
```
linux> strace ./hello
execve("./hello", ["hello"], [/* ... */]).
...
write(1, "hello\n", 6)                  = 6
...
exit_group(0)                           = ?
```

Today

- I/O Systems
- Unix I/O
- Metadata, sharing, and redirection
- Standard I/O
- **Closing remarks**

Standard I/O vs. Unix I/O

- Standard I/O is implemented using low-level Unix I/O



- Which ones should you use in your programs?

用户I/O软件

用户软件可用以下两种方式提出I/O请求：

（1）使用高级语言提供的标准I/O库函数。例如，在C语言程序中可以直接使用像fopen、fread、fwrite和fclose等文件操作函数，或printf、putc、scanf和getc等控制台I/O函数。 程序移植性很好！

但是，使用标准I/O库函数有以下几个方面的不足：

(a) 标准I/O库函数不能保证文件的安全性（无加/解锁机制）

(b) 所有I/O都是同步的，程序必须等待I/O操作完成后才能继续执行

(c) 有时不适合甚至无法使用标准I/O库函数实现I/O功能，如，不提供读取文件元数据的函数（元数据包括文件大小和文件创建时间等）

(d) 用它进行网络编程会造成易于出现缓冲区溢出等风险

（2）使用OS提供的API函数或系统调用。例如，在Windows中直接使用像CreateFile、ReadFile、WriteFile、CloseHandle等文件操作API函数，或ReadConsole、WriteConsole等控制台I/O的API函数。对于Unix或Linux用户程序，则直接使用像open、read、write、close等系统调用封装函数。

Pros and Cons of Standard I/O

- **Pros:**
  - Buffering increases efficiency by decreasing the number of **`read`** and **`write`** system calls
  - Short counts are handled automatically
- **Cons:**
  - Provides no function for accessing file metadata
  - Standard I/O functions are not async-signal-safe, and not appropriate for signal handlers
  - Standard I/O is not appropriate for input and output on network sockets
    - There are poorly documented restrictions on streams that interact badly with restrictions on sockets (CS:APP3e, Sec 10.11)

系统I/O软件

OS在I/O子系统中的重要性由I/O系统以下三个特性决定：

（1）共享性。I/O系统被多个程序共享，须由OS对I/O资源统一调度管理，以保证用户程序只能访问自己有权访问的那部分I/O设备，并使系统的吞吐率达到最佳。

（2）复杂性。I/O设备控制细节复杂，需OS提供专门的驱动程序进行控制，这样可对用户程序屏蔽设备控制的细节。

（3）异步性。不同设备之间速度相差较大，因而，I/O设备与主机之间的信息交换使用异步的中断I/O方式，中断导致从用户态向内核态转移，因此必须由OS提供中断服务程序来处理。

那么，如何从用户程序对应的用户进程进入到操作系统内核执行呢？

系统调用！

Pros and Cons of Unix I/O

- **Pros**
  - Unix I/O is the most general and lowest overhead form of I/O
    - All other I/O packages are implemented using Unix I/O functions
  - Unix I/O provides functions for accessing file metadata
  - Unix I/O functions are async-signal-safe and can be used safely in signal handlers

- **Cons**
  - Dealing with short counts is tricky and error prone
  - Efficient reading of text lines requires some form of buffering, also tricky and error prone
  - Both of these issues are addressed by the standard I/O

系统调用和API

- OS提供一组系统调用为用户进程的I/O请求进行具体的I/O操作。
- 应用编程接口（API）与系统调用两者在概念上不完全相同，它们都是系统提供给用户程序使用的编程接口，但前者指的是功能更广泛、抽象程度更高的函数，后者仅指通过软中断（自陷）指令向内核态发出特定服务请求的函数。
- 系统调用封装函数是 API 函数中的一种。
- API 函数最终通过调用系统调用实现 I/O。一个API 可能调用多个系统调用，不同 API 可能会调用同一个系统调用。但是，并不是所有 API 都需要调用系统调用。
- 从编程者来看，API 和 系统调用之间没有什么差别。
- 从内核设计者来看，API 和 系统调用差别很大。API 在用户态执行，系统调用封装函数也在用户态执行，但具体服务例程在内核态执行。

Choosing I/O Functions

- **General rule: use the highest-level I/O functions you can**
  - Many C programmers are able to do all of their work using the standard I/O functions
  - But, be sure to understand the functions you use!

- **When to use standard I/O**
  - When working with disk or terminal files
- **When to use raw Unix I/O**
  - *Inside signal handlers, because Unix I/O is async-signal-safe*
  - In rare cases when you need absolute highest performance

Aside: Working with Binary Files

- **Functions you should *never* use on binary files**
  - **Text-oriented I/O:** such as `fgets, scanf`
    - Interpret EOL characters.
  - **String functions**
    - `strlen, strcpy, strcat`
    - Interprets byte value 0 (end of string) as special

How does the CPU talk to devices?

- **Device controller: Hardware that enables devices to talk to the peripheral bus**
- **Host adapter: Hardware that enables the computer to talk to the peripheral**
- **Bus: Wires that transfer data between components inside computer**
- **Device controller: allows OS to specify simpler instructions to access data**
- **Example: a disk controller**
  - Translates “access sector 23” to “move head reader 1.672725272 cm from edge of platter”
  - Disk controller “advertises” disk parameters to OS, hides internal disk geometry. Most modern hard drives have disk controller embedded as a chip on the physical device

I/O Hardware

- **Incredible variety of I/O devices**
  - Storage
  - Transmission
  - Human-interface
- **Common concepts – signals from I/O devices interface with computer**
  - **Port** – connection point for device
  - **Bus** - daisy chain or shared direct access
  - **Controller** (**host adapter**) – electronics that operate port, bus, device
    - Sometimes integrated
    - Sometimes separate circuit board (host adapter)
    - Contains processor, microcode, private memory, bus controller, etc
      - Some talk to per-device controller with bus controller, microcode, memory, etc

I/O Hardware (Cont.)

- **I/O instructions control devices**
- **Devices usually have registers where device driver places commands, addresses, and data to write, or read data from registers after command execution**
  - Data-in register, data-out register, status register, control register
  - Typically 1-4 bytes, or FIFO buffer
- **Devices have addresses, used by**
  - Direct I/O instructions
  - **Memory-mapped I/O**
    - Device data and command registers mapped to processor address space
    - Especially for large address spaces (graphics)

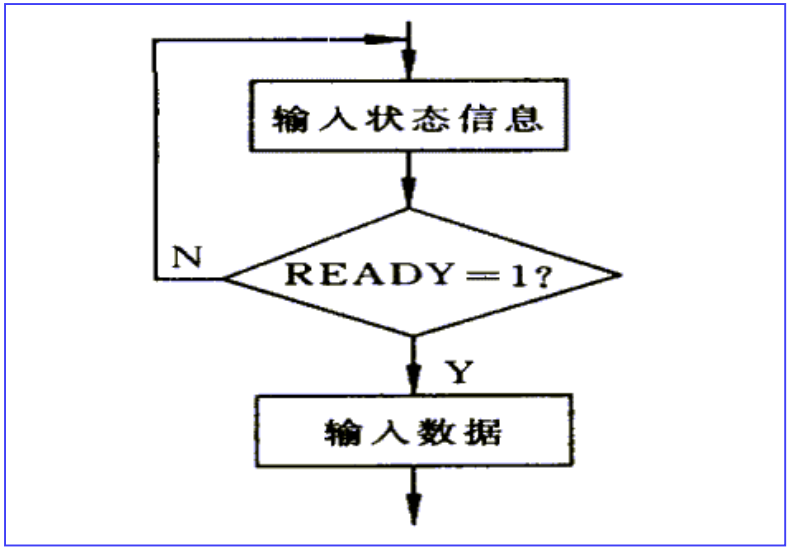
Three Types of I/O

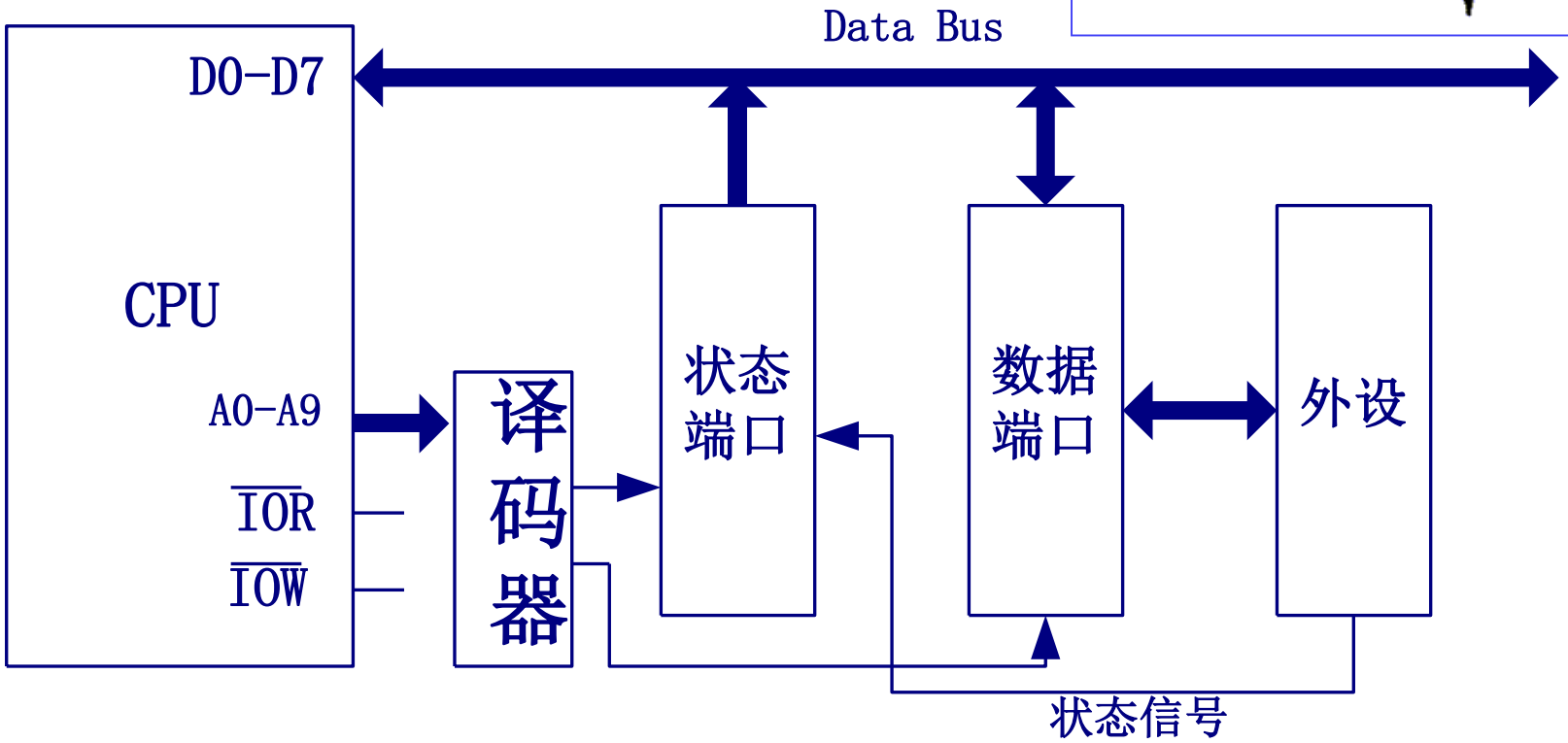
- **Programmed I/O(Polling): continuous attention of the processor is required**
  - 无条件传统方式
  - 查询方式

- **Interrupt driven I/O: processor launches I/O and can continue until interrupted**

- **Direct memory access(DMA): the dma module governs the exchange of data between the I/O unit and the main memory**

查询方式


输入状态信息
N
READY＝1?
Y
输入数据


Data Bus
D0-D7
CPU
A0-A9
IOR
IOW
译码器
状态端口
数据端口
外设
状态信号

程序查询（Polling）方式

- I/O设备（包括设备控制器）将自己的状态放到状态寄存器中
  - 打印缺纸、打印机忙、未就绪等都是状态
- OS阶段性地查询状态寄存器中的特定状态，以决定下一步动作
  - 如：未“就绪”时，则一直“等待”
- 例如：sys_write进行字符串打印的程序段大致过程如下：

```
copy_string_to_kernel ( strbuf, kernelbuf, n);  // 将字符串复制到内核缓冲区
for (i=0; i < n; i++) {                         // 对于每个打印字符循环执行
  while ( printer_status != READY);             // 等待直到打印机状态为“就绪”
  *printer_data_port=kernelbuf[i];              // 向数据端口输出一个字符
  *printer_control_port=START;                  // 发送“启动打印”命令
}
return_to_user ( );                             // 返回用户态
```

如何判断“就绪”？如何“等待”？

读取状态寄存器，判断特定位（1-就绪；0-未就绪）是否为1

等待：读状态、判断是否为1；不是，则继续读状态、判断、……

打印输出标准子程序

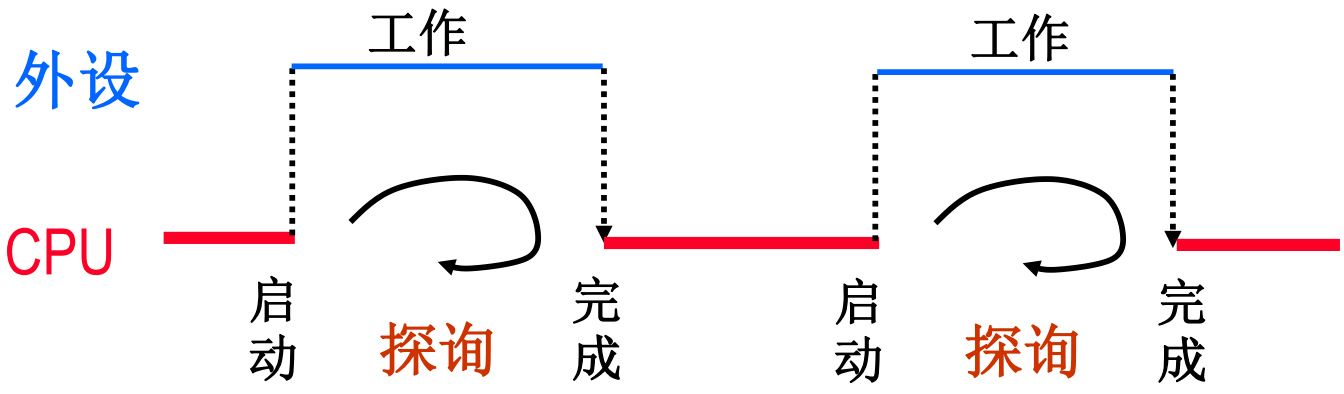
功能：打印AL寄存器中的字符。

```
PRINT     PROC  NEAR
          PUSH  AX          ; 保留用到的寄存器
          PUSH  DX          ; 保留用到的寄存器
          MOV   DX, 378H    ; 数据锁存器口地址送DX
          OUT   DX, AL      ; 输出要打印的字符到数据锁存器
          MOV   DX, 379H    ; 状态寄存器口地址送DX
WAIT:     IN    AL, DX      ; 读打印机状态位
          TEST  AL, 80H     ; 检查忙位
          JE    WAIT        ; 等待直到打印机不忙
          MOV   DX, 37AH    ; 命令(控制)寄存器口地址送DX
          MOV   AL, 0DH     ; 置选通位=1（表示启动打印）
          OUT   DX, AL      ;使命令寄存器中选通位置1
          POP   DX
          POP   AX          ; 恢复寄存器
          RET
PRINT     ENDP
```

回顾：过程/函数/子程序中的开始总是先要保护现场，最后总是要恢复现场！

程序查询I/O方式

sys_write系统调用服务例程



“踏步”现象

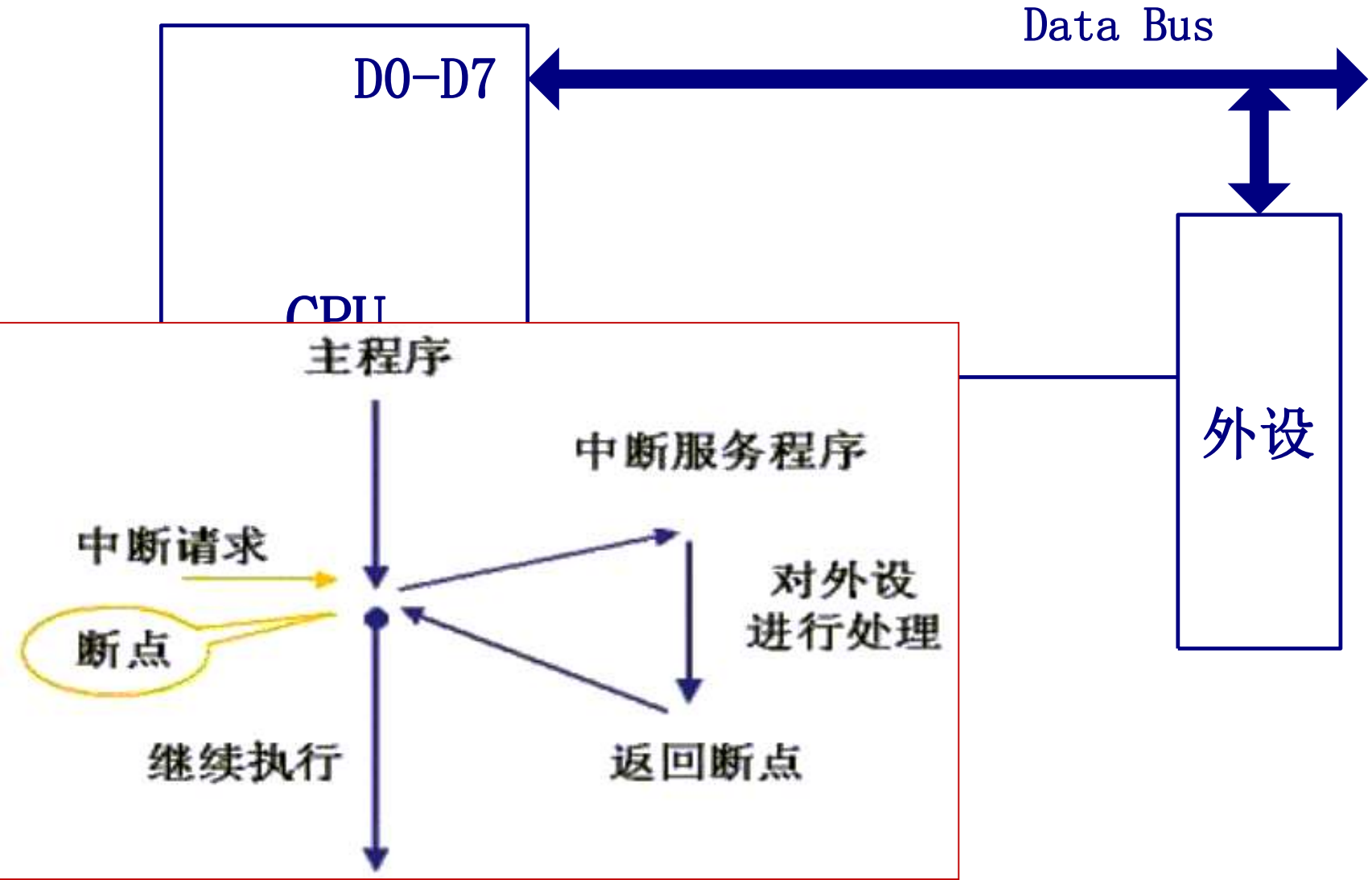
此时，CPU处于停止状态吗？

不是！只是不断执行 “ IN-TEST-JE”3条指令，称为“忙等待”！

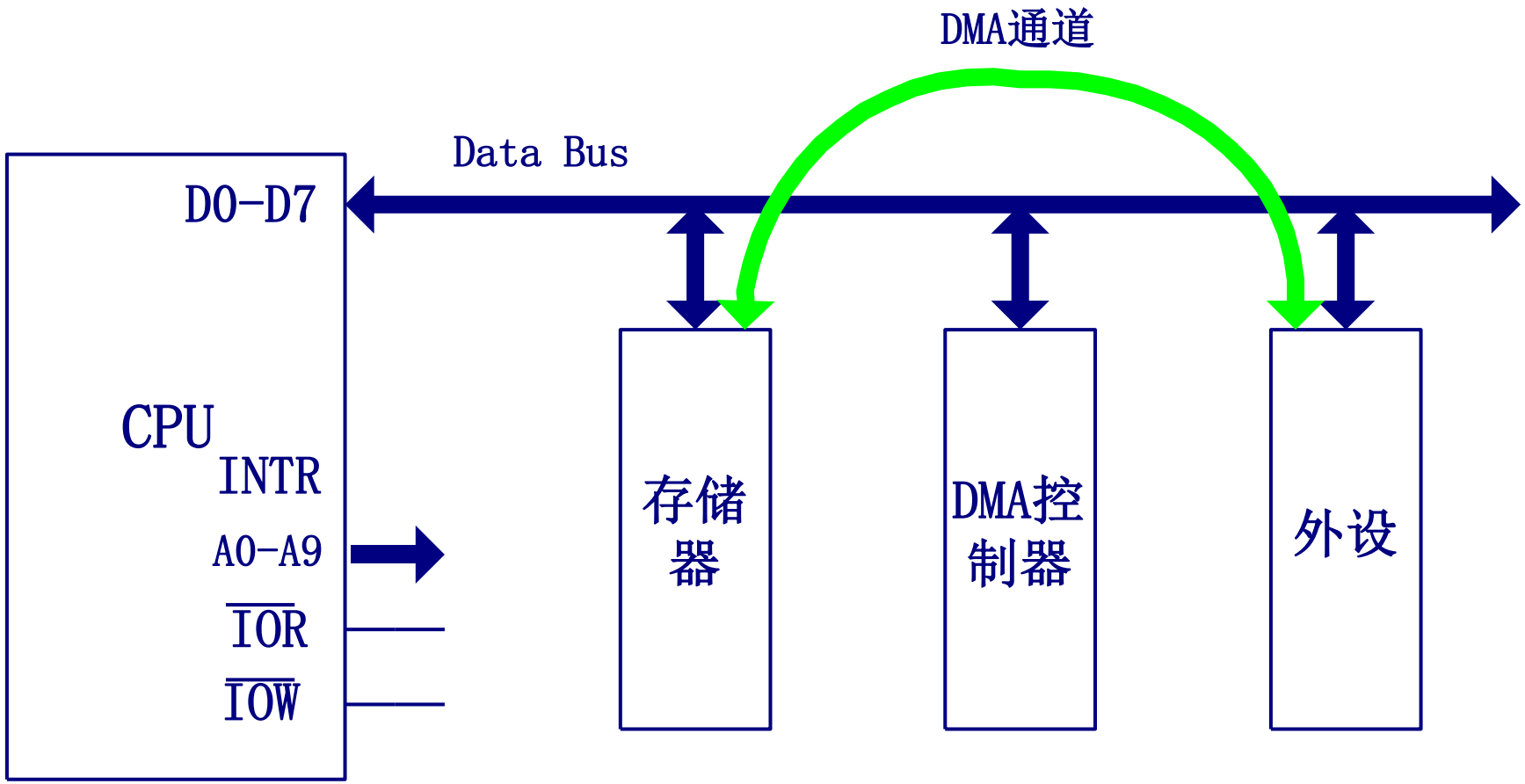
“探询”期间，可一直不断查询（独占查询），也可定时查询（需保证数据不丢失！）。

- **特点：**
  - 简单、易控制、外围接口控制逻辑少；
  - CPU与外设串行工作，效率低、速度慢，适合于慢速设备
  - 查询开销极大 (CPU完全在等待“外设完成”)
- **工作方式：**完全串行或部分串行，CPU用100%的时间为I/O服务！

中断方式



DMA方式



Extra Slides

Fun with File Descriptors (1)

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1, fd2, fd3;
    char c1, c2, c3;
    char *fname = argv[1];
    fd1 = Open(fname, O_RDONLY, 0);
    fd2 = Open(fname, O_RDONLY, 0);
    fd3 = Open(fname, O_RDONLY, 0);
    Dup2(fd2, fd3);
    Read(fd1, &c1, 1);
    Read(fd2, &c2, 1);
    Read(fd3, &c3, 1);
    printf("c1 = %c, c2 = %c, c3 = %c\n", c1, c2, c3);
    return 0;
}
```

ffiles1.c

- **What would this program print for file containing "abcde"?**

Fun with File Descriptors (2)

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1;
    int s = getpid() & 0x1;
    char c1, c2;
    char *fname = argv[1];
    fd1 = Open(fname, O_RDONLY, 0);
    Read(fd1, &c1, 1);
    if (fork()) { /* Parent */
        sleep(s);
        Read(fd1, &c2, 1);
        printf("Parent: c1 = %c, c2 = %c\n", c1, c2);
    } else { /* Child */
        sleep(1-s);
        Read(fd1, &c2, 1);
        printf("Child: c1 = %c, c2 = %c\n", c1, c2);
    }
    return 0;
}
```

ffiles2.c

- **What would this program print for file containing "abcde"?**

Fun with File Descriptors (3)

```
#include "csapp.h"
int main(int argc, char *argv[])
{
    int fd1, fd2, fd3;
    char *fname = argv[1];
    fd1 = Open(fname, O_CREAT|O_TRUNC|O_RDWR, S_IRUSR|S_IWUSR);
    Write(fd1, "pqrs", 4);
    fd3 = Open(fname, O_APPEND|O_WRONLY, 0);
    Write(fd3, "jklmn", 5);
    fd2 = dup(fd1);  /* Allocates descriptor */
    Write(fd2, "wxyz", 4);
    Write(fd3, "ef", 2);
    return 0;
}
```

ffiles3.c

- **What would be the contents of the resulting file?**

Accessing Directories

- **Only recommended operation on a directory: read its entries**
  - **`dirent`** structure contains information about a directory entry
  - DIR structure contains information about directory while stepping through its entries

```
#include <sys/types.h>
#include <dirent.h>

{
  DIR *directory;
  struct dirent *de;
  ...
  if (!(directory = opendir(dir_name)))
      error("Failed to open directory");
  ...
  while (0 != (de = readdir(directory))) {
      printf("Found file: %s\n", de->d_name);
  }
  ...
  closedir(directory);
}
```

Example of Accessing File Metadata

```
int main (int argc, char **argv)
{
    struct stat stat;
    char *type, *readok;

    Stat(argv[1], &stat);
    if (S_ISREG(stat.st_mode))      /* Determine file type */
        type = "regular";
    else if (S_ISDIR(stat.st_mode))
        type = "directory";
    else
         type = "other";
    if ((stat.st_mode & S_IRUSR)) /* Check read access */
        readok = "yes";
    else
         readok = "no";

    printf("type: %s, read: %s\n", type, readok);
    exit(0);
}
```

statcheck.c

```
linux> ./statcheck statcheck.c
type: regular, read: yes
linux> chmod 000 statcheck.c
linux> ./statcheck statcheck.c
type: regular, read: no
linux> ./statcheck ..
type: directory, read: yes
```

本章小结

- 用户程序通常通过调用编程语言提供的库函数或操作系统提供的API函数来实现I/O操作
- I/O库函数最终都会调用系统调用的封装函数，通过封装函数中的陷阱指令使用户进程从用户态转到内核态执行
- 在内核态中执行的内核空间I/O软件主要包含三个层次：
  - 与设备无关的操作系统软件
  - 设备驱动程序
  - 中断服务程序
- 具体I/O操作通过设备驱动程序和中断服务程序控制I/O硬件来实现
- 设备驱动程序的实现主要取决于具体的I/O控制方式：
  - 程序查询方式、中断方式、DMA方式